

Daily Geometry Drill #1

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Gradient, intercept, midpoint, distance — read them straight off. No working shown needed.

- Find the gradient of the line $6x - 3y + 15 = 0$.
- Find the gradient of the line through $(1, 2)$ and $(5, 14)$.
- A line of gradient 4 is perpendicular to a line of gradient m . Write down m .
- Find the distance between $(2, 1)$ and $(5, 5)$.
- Find the midpoint of the segment joining $(4, 8)$ and $(10, 2)$.
- Find the perpendicular distance from the origin to the line $3x + 4y - 10 = 0$.
- A line has x -intercept 4 and y -intercept 6. Find its gradient.
- Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$.
- The points $(1, 2)$, $(2, 4)$ and $(k, 10)$ are collinear. Find k .
- Using the shoelace formula, find the area of the triangle with vertices $(0, 0)$, $(3, 4)$ and $(6, 0)$.

Section B Manipulation Drills

(≤50 s each)

Build the line, then interrogate it. Extra tool required per question.

- The line $2x - 3y + 6 = 0$ is perpendicular to which of the following?

A) $y = -\frac{3}{2}x + 1$

B) $y = \frac{3}{2}x - 2$

C) $y = \frac{2}{3}x + 5$

D) $y = -\frac{2}{3}x + 3$

2. What is the perpendicular distance from $(2, 3)$ to the line $5x + 12y - 7 = 0$?

A) 1

B) 2

C) 3

D) 4

3. Which point lies on the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$?

A) $(2, 5)$

B) $(3, 3)$

C) $(4, 4)$

D) $(5, 3)$

4. Find the area of the quadrilateral with vertices $(0, 0)$, $(4, 0)$, $(5, 3)$, $(1, 3)$, in that order.

A) 10

B) 12

C) 14

D) 16

5. The line through $(1, 2)$ and $(4, 5)$ crosses the y -axis at which value?

A) 0

B) 1

C) 2

D) 3

6. Find the perpendicular distance from $(0, 1)$ to the line $6x - 8y + 2 = 0$.

7. Which of the following is NOT parallel to $y = 3x + 1$?

A) $y = 3x + 5$

B) $6x - 2y = 4$

C) $3x - y = 2$

D) $y = 2x + 1$

8. The line $y = x + 1$ meets the line through $(0, 3)$ perpendicular to it at the point P . Find the coordinates of P .

9. Which point is closest to the origin?

- A) (1, 3)
- B) (2, 2)
- C) (4, 0)
- D) (0, 5)

10. Using the shoelace formula, find the area of the triangle with vertices (1, 1), (4, 2), (2, 6).

Section C Substitution & Structure

(≤90 s each)

Spot the tool first — the numbers were chosen so that a clean route exists.

1. The midpoint of (2, -1) and (6, 7) is?
 - A) (4, 3)
 - B) (3, 4)
 - C) (5, 4)
 - D) (4, 2)
2. The perpendicular bisector of the segment joining (0, 1) and (4, 5) has equation?
 - A) $x + y = 5$
 - B) $y = x + 1$
 - C) $y = -x + 3$
 - D) $y = x + 3$
3. The distance between (2, 5) and (7, 17) is?
 - A) 13
 - B) 14
 - C) $\sqrt{194}$
 - D) 12
4. The area of the triangle with vertices (0, 0), (5, 0), (2, 6) is?
 - A) 15
 - B) 30
 - C) 10
 - D) 12
5. Which line is perpendicular to $y = 2x - 3$?
 - A) $y = -2x + 1$
 - B) $y = -\frac{1}{2}x + 4$
 - C) $y = \frac{1}{2}x$

D) $y = 2x + 5$

6. The points $(1, 4)$, $(4, 7)$, $(10, k)$ are collinear. Find k .

A) 10

B) 12

C) 13

D) 15

7. The perpendicular distance from $(2, -1)$ to the line $3x - 4y + 10 = 0$ is?

A) 2

B) 3

C) 4

D) 5

8. The line through $(2, 1)$ perpendicular to $y = \frac{1}{2}x + 3$ has equation?

A) $y = -2x + 5$

B) $y = 2x - 3$

C) $y = -\frac{1}{2}x + 1$

D) $y = -2x + 3$

Section D Challenge

(TMUA / SMC difficulty)

Synthesis: two skills at once, then check your short answer against the options.

1. The perpendicular bisector of the segment joining $(3, -1)$ and $(7, 5)$ meets the x -axis at P . What is the x -coordinate of P ?

A) 6

B) 7

C) 8

D) 9

2. A triangle with vertices $(0, 0)$, $(6, 0)$ and $(3, h)$ has area 12. Find h .

A) 2

B) 4

C) 6

D) 8

3. The distance from the point $(4, 3)$ to the line $y = x$ is?

A) $\frac{1}{\sqrt{2}}$

- B) 1
- C) $\sqrt{2}$
- D) $\frac{7}{\sqrt{2}}$

4. The line $2x + 3y - 12 = 0$ cuts the coordinate axes to form a right-angled triangle. Its area is?
- A) 6
 - B) 8
 - C) 12
 - D) 24
5. Using the shoelace formula, find the area of the quadrilateral with vertices $(0, 2)$, $(3, 0)$, $(6, 4)$, $(1, 6)$, in that order.

“Shoelace before altitude: area is a one-line cross product, not a construction.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths